

Romeo Silvestri

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Summary

Machine Learning Engineer with 3+ years of professional and research experience, combining a strong foundation in statistics and data science with applied machine learning and AI research. Experienced in developing end-to-end ML systems across forecasting, reinforcement learning, and generative AI, from scientific experimentation and validation to MLOps and production deployment. **Author of peer-reviewed publications**, with a focus on translating research into robust, data-driven systems with real-world applications.

Technical Skills

Programming Languages: Python, R, SQL

Machine Learning & Statistics: Deep Learning (PyTorch, TensorFlow), Reinforcement Learning (Gymnasium), Time Series Forecasting, Statistical Modeling (statsmodels), Natural Language Processing (NLP), Anomaly & Drift Detection, Classification, Regression & Clustering (scikit-learn), Hyperparameter Tuning (Optuna), A/B Testing

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph

MLOps & Engineering: MLflow, Docker, FastAPI, Airflow, Prefect, Model Monitoring, Git, uv

Data & Analytics: Pandas, NumPy, Spark (PySpark), PostgreSQL, MySQL, MongoDB, Power BI

Work Experience

ML Engineer | Fondazione Bruno Kessler (FBK), Trento

12/2023 – Present

- Designed, tested, and deployed a Decision Support System for water management, integrating heterogeneous data sources, including IoT sensors and weather data, reducing operational costs by around **50%** across several companies.
- Operationalized research prototypes through reproducible ML pipelines and APIs using Python, FastAPI, MLflow, and GitLab, supporting experimentation and production deployment in collaboration with domain experts.
- Developed a PPO-based Deep Reinforcement Learning agent in a digital-twin environment combining surrogate and physics-based models for simulation and policy training.
- Developed **WeatherSynth**, a modular Python library for composing and benchmarking reproducible synthetic weather-generation pipelines with interchangeable generators, climate conditioners, physical constraints, and diagnostics.
- Developed a multi-agent LLM/RAG assistant using LangChain and LangGraph, integrating technical and regulatory knowledge for real-time agricultural decision support with global and region-specific recommendations.
- Built and evaluated ARIMA, XGBoost, and LSTM models for sensor time-series forecasting, achieving low prediction error through feature engineering, hyperparameter tuning, and rigorous validation.
- Developed and benchmarked a Wasserstein-based drift detector for online regression monitoring, comparing it with established methods and evaluating its applicability to sensor forecasting.
- Evaluated IoT soil-sensor behavior and anomaly-detection methods in collaboration with industry and academic partners to assess sensor quality, reliability, and suitability for practical field use.

Data Scientist | Youbidata, Bologna

10/2023 – 11/2023

- Developed a centralized financial monitoring dashboard using PostgreSQL and Power BI to consolidate, track, and visualize key financial KPIs, supporting reporting and decision-making for a consulting client.

Statistician (Internship) | ISTAT, Rome

03/2023 – 08/2023

- Analyzed national demographic data using R and SQL to uncover socio-economic trends and produce periodic reports.
- Evaluated the Austrian quality framework's suitability for the Italian Census, achieving statistically significant results.

Selected Projects

IT4LIA AI Factory | EuroHPC Joint Undertaking

04/2025 – Present

Contributing to the Agrifood vertical, supporting the development and delivery of AI services for research and industry.

IRRITRE | Autonomous Province of Trento / FBK

12/2023 – Present

Contributing to a regional precision-irrigation initiative combining digital tools and data for sustainable water management.

AgrifoodTEF | Digital Europe Programme

09/2025 – 12/2025

Contributed to the testing and validation of IoT soil-sensing technologies in collaboration with an industry sensor provider.

Education

M.Sc. in Statistics and Data Science | Sapienza University of Rome

11/2020 – 03/2023

Thesis: *Comparison of Forecasting Models for the Real Estate Market: Application to Property Prices in Madrid*

Final grade: 107/110

B.Sc. in Statistics | University of Padua

10/2017 – 11/2020

Thesis: *The Effect of Migration on Demographic Dynamics: Application to Italian Provinces*

Final grade: 102/110

Publications

Silvestri, R., Pincheira, M., Vecchio, M., & Antonelli, F. A Retrieval-Augmented LLM Framework for PRISMA-Aligned Systematic Reviews. *Neural Computing and Applications*, **accepted for publication**, 2026.

Silvestri, R., Antonini, M., Vecchio, M., & Antonelli, F. Deep Reinforcement Learning for Irrigation Optimization Based on Crop-Soil Dynamics. *IEEE International Workshop on Metrology for Industry 4.0 & IoT (MetroInd4.0 & IoT)*, 2026.

Silvestri, R., Vecchio, M., Pincheira, M., & Antonelli, F. Smart Irrigation with Fuzzy Decision Support Systems in Trentino Vineyards. *Sensors*, 2025.

Silvestri, R., Vecchio, M., & Antonelli, F. A Fuzzy Decision Support System to Optimize Irrigation Practices in Trentino Region. *IEEE International Conference on Control, Decision and Information Technologies (CoDIT)*, 2025.

Silvestri, R., Vecchio, M., Pincheira, M., & Antonelli, F. Comparative Analysis of Soil Moisture Interpolation Techniques in Apple Orchards of Trentino Region. *IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor)*, 2024.

Grazieschi, P., **Silvestri, R.**, Antonini, M., & Vecchio, M. A Calibrated Wasserstein Drift Detector for Residual-Based Monitoring of Regression Models. *Neurocomputing*, **under review**, 2026.

Additional Information

Teaching: University Tutor, Sapienza University of Rome (2022); Private Tutor in Statistics and IT (2020–2021)

Languages: Italian (Native); English (B2+, Professional Working Proficiency)

Certifications: Deep Learning (DeepLearning.AI); Google Data Analytics; IBM Data Science

Memberships & Service: Volunteer Blood Donor, AVIS; Mensa Italia

Interests: Art, Chess, Finance, Football, Mixed Martial Arts, Music, Poetry, Travel